ASSIGNMENT 3

Q7 - Smart Calculator

JavaScript solution

```
// q7.js
"use strict";

class InvalidOperationError extends Error {
  constructor(operation) {
    super(`Unknown operation: ${operation}`);
    this.name = "InvalidOperationError";
  }
}

function calculate(operation, num1, num2) {
  switch (operation) {
    case "add": return num1 + num2;
    case "subtract": return num1 - num2;
    case "divide":
      if (num2 === 0) throw new Error("Cannot divide by zero");
      return num1 / num2;
    case "power": return num1 ** num2;
    case "root":
      if (num1 < 0) throw new Error("Cannot take a real square root of a negative number");
      return Math.sqrt(num1);
    default: throw new InvalidOperationError(operation);
  }
}

const operations = ["add", "divide", "power", "root", "subtract"];
const num1 = 25, num2 = 0;

for (const operation of operations) {
  try {
    const result = calculate(operation, num1, num2);
    console.log(`${operation.toUpperCase()}: ${result}`);
  } catch (error) {
    console.log(`${operation.toUpperCase()} ERROR: ${error.name} - ${error.message}`);
  }
}
```

Explanation / Debug notes

The switch selects an operation. Division by zero and square root of a negative number throw errors. The default case throws InvalidOperationError for unknown operations. Each operation is wrapped in try...catch so one error does not prevent later operations.

Run the code in the browser console or Node.js. Save each solution in its own .js file as required.